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Mid-Term Project: Numerical experiments on quantum relaxation

1. Introduction

Relaxation processes play an important role in many areas of chemistry. When a discrete quantum state is coupled to a large number of surrounding states, probability can flow irreversibly from the initial state into the surrounding manifold. This process is closely related to phenomena like spontaneous emission, molecular predissociation, and energy transfer in condensed phase systems. When the surrounding states form a dense spectrum, the probability of remaining in the initial state often decays approximately exponentially, a behavior that can be understood through Fermi's Golden Rule.

In this project, numerical simulations are performed to study the time evolution of quantum systems that consist of discrete states coupled to a manifold of closely spaced energy levels. Two models are investigated: in Model (A), a single discrete state is coupled directly to a continuum of levels; and in Model (B), an additional intermediate state is introduced so that population transfer occurs first between two discrete states before relaxation into the continuum. These models provide a simple framework for exploring how the structure of the Hamiltonian and the distribution of coupling strengths influence relaxation behavior.

The goal of the numerical experiment is to compute the time evolution of the system starting from a specified initial state and to evaluate the probability of finding the system in different basis states as a function of time. In particular, the probability of remaining in the initial state and the probability of occupying the continuum manifold are examined. The dependence of the results on parameters such as the bandwidth of the continuum, the number of states, and the coupling strengths is also investigated.

2. Theory / Method (from lecture)

2.1 Time-dependent Schrödinger equation

The dynamics of the quantum system are governed by the time-dependent Schrodinger equation:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = \hat{H}|\psi(t)\rangle$$

where $\hat{H}$ is the Hamiltonian operator and $|\psi(t)\rangle$ is the state vector of the system.

For a time-independent Hamiltonian, the formal solution of this equation can be written in terms of the time evolution operator:

$$|\psi(t)\rangle = e^{-i\hat{H}t/\hbar}|\psi(0)\rangle$$

where $|\psi(0)\rangle$ is the initial state of the system.

In order to evaluate the time evolution numerically, the Hamiltonian is represented as a matrix in a discrete basis of states. The Hamiltonian matrix is diagonalized so that:

$$H = UDU^{\dagger}$$

where D is a diagonal matrix containing the eigenvalues of the Hamiltonian and U is the matrix whose columns are the corresponding eigenvectors. Using this docomposition, the time evolution operator can be written as:

$$e^{-i\hat{H}t/\hbar} = Ue^{-iDt/\hbar}U^{\dagger}$$

where

$$e^{-iDt/\hbar} = diag(e^{-iE_{\alpha}t/\hbar})$$

with E_{α} the eigenvalues of the Hamiltonian. Then the state vector at time t can therefore be evaluated as:

$$|\psi(t)\rangle = Ue^{-iDt/\hbar}U^{\dagger}|\psi(0)\rangle$$

In the numerical calculations following, units are chosen such that $\hbar = 1$, which simplifies the expressions above.

2.2 Expansion in the basis states

The time-dependent wavefunction can be expanded in the chosen basis states of the Hamiltonian. For Model (A), the basis consists of a single discrete state $|1\rangle$ and a manifold of states $|k\rangle$:

$$|\psi(t)\rangle = C_1(t)|1\rangle + \sum_{k=1}^{N} C_k(t)\,|k\rangle$$

The coefficients $C_n(t)$ represents the probability amplitudes for the system to occupy the corresponding states.

For Model (B), the basis includes two discrete states and the continuum manifold:

$$|\psi(t)\rangle = C_0(t)|0\rangle + C_1(t)|1\rangle + \sum_{k=1}^{N} C_k(t)\,|k\rangle$$

The initial conditions for the two models are chosen according to the problem statement. In Model (A), the system begins in the discrete state $C_1(0) = 1$, $C_k(0) = 0$. In Model (B), the initial state is $C_0(0) = 1$, $C_1(0) = 0$, $C_k(0) = 0$.

2.3 Hamiltonian models

Model A:

In Model (A), a single discrete state $|1\rangle$ is coupled to a set of N continuum states $|k\rangle$. The Hamiltonian is:

$$\hat{H} = E_1|1\rangle\langle 1| + \sum_{k=1}^{N} E_k|k\rangle\langle k| + \sum_{k=1}^{N}(V_{1k}|1\rangle\langle k| + V_{k1}|k\rangle\langle 1)$$

This model describes the decay of a discrete state into a large set of surrounding states.

Model B:

In Model (B), an additional state $|0\rangle$ is coupled to $|1\rangle$, while the state $|1\rangle$ remains coupled to the continuum. The Hamiltonian is:

$$\hat{H} = E_0|0\rangle\langle 0| + E_1|1\rangle\langle 1| + V_{01}|0\rangle\langle 1| + V_{10}|1\rangle\langle 0| + \sum_{k=1}^{N} E_k|k\rangle\langle k| + \sum_{k=1}^{N}(V_{1k}|1\rangle\langle k| + V_{k1}|k\rangle\langle 1)$$

This model allows population transfer between two discrete states followed by relaxation into the continuum.

2.4 Continuum energy distribution

The continuum states are distributed over an energy band of width ΔE centered around E_1. The energies of the continuum states are taken to be equally spaces:

$$E_k \in \left[E_1 - \frac{\Delta E}{2}, E_1 + \frac{\Delta E}{2}\right]$$

If N states are used to represent the continuum, the spacing between neighboring levels is:

$$\epsilon = \frac{\Delta E}{N}$$

The density of states is therefore:

$$\rho = \frac{1}{\epsilon} = \frac{N}{\Delta E}$$

2.5 Calculation of probabilities

The probability of finding the system in a basis state $|b\rangle$ at time t is given by:

$$P_b(t) = |\langle b|\psi(t)\rangle|^2$$

Because the wavefunction is expanded in the basis states, this corresponds directly to the magnitude square of the corresponding coefficient:

$$P_b(t) = |C_b(t)|^2$$

For Model (A), the probability of remaining in the initial state is:

$$P_1(t) = |C_1(t)|^2$$

The total probability of occupying the continuum is:

$$P_K(t) = \sum_{k=1}^{N} |C_k(t)|^2$$

For Model (B), the probabilities of occupying the discrete states are:

$$P_0(t) = |C_0(t)|^2$$

$$P_1(t) = |C_1(t)|^2$$

and the probability of occupying the continuum is agiain:

$$P_K(t) = \sum_{k=1}^{N} |C_k(t)|^2$$

3. Numerical setup

The numerical simulations were carried out in Python using the NumPy and SciPy library for numerical linear algebra and Matplotlib for visualization.

3.1 Construction of the Hamiltonian matrix

For Model (A), the basis states are: $|1\rangle$, $|k_1\rangle$, $|k_2\rangle$, . . . $|k_N\rangle$. The Hamiltonian matrix therefore has dimension $(N+1)\times(N+1)$. The diagonal elements correspond to the enegies of the states $H_{11}=E_1$, $H_{kk}=E_k$, while the off-diagonal elements describe the coupling between the discrete state and the continuum states, $H_{1k}=V_{1k}$, $H_{k1}=V_{k1}$.

For Model (B), the basis is extended to include the additional state $|0\rangle$, which is $|0\rangle$, $|1\rangle$, $|k_1\rangle$, $|k_2\rangle$, . . . $|k_N\rangle$, giving a Hamiltonian matrix of dimension $(N+2)\times(N+2)$. In addition to the couplings between $|1\rangle$ and the continuum states, the Hamiltonian includes the interaction between the two discrete states $H_{01}=V_{01}$, $H_{10}=V_{10}$. The Hamiltonians matrices were constructed numerically so that H is Hermitian.

3.2 Continuum energy distribution

The continuum was represented by a finite set of N states whose energies were uniformly distributed over a band of width ΔE centered around E_1. The energies were defined as:

$$E_k = E_1 - \frac{\Delta E}{2} + k\epsilon$$

where $\epsilon = \frac{\Delta E}{N}$. A value of $N = 150$ was used to approximate a dense spectrum of states. Without loss of generality, the energy of state $|1\rangle$ was set to $E_1 = 0$.

3.3 Numerical time evolution

The Hamiltonian matrix was diagonalized using 'numpy.linalg.eigh', which computes the eigenvalues and eigenvectors of Hermitian matrices. As stated in the sections above, if the Hamiltonian is written as $H = UDU^\dagger$, where D contains the eigenvalues and U the eigenvectors, the time evolution operator can be evaluated as $U(t) = e^{-iHt} = Ue^{-iDt}U^\dagger$. The time-dependent wavefunction was then obtained from $|\psi(t)\rangle = U(t)|\psi(0)\rangle$. The state vector was evaluated at a sequence of time points t_i spanning the duration of the simulation.

3.4 Calculation of probabilities

The coefficients of the wavefunction expansion correspond directly to the probability of amplitudes of the basis sets, as in equations provided above. The probabilities were evaluated for each time point and plotted as functions of time to analyze the relaxation.

4. Results for Model (A)

4.1 Time Evolution of State Populations

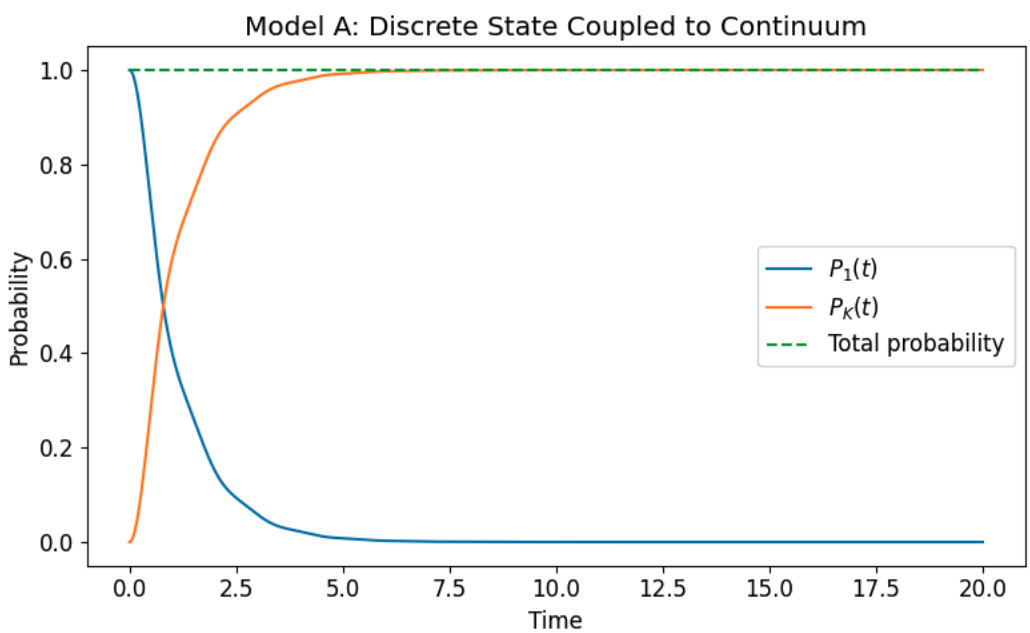

Figure 1. Time evolution of the survival probability $P_1(t)$ and the continuum population $P_k(t)$ for Model (A) with $N = 150$, $\Delta E = 10$, and $V = 0.1$. The total probability is also shown to confirm normalization.

Shown in **Figure 1** above, the survival probability $P_1(t)$ decreases monotonically while the continuum population $P_k(t)$ increases correspondingly, indicating transfer of probability from the discrete state into the continuum. The total probability remains equal to unity at all times, confirming the unitary nature of the time evolution. The smooth decay of $P_1(t)$ suggests relaxation into the continuum.

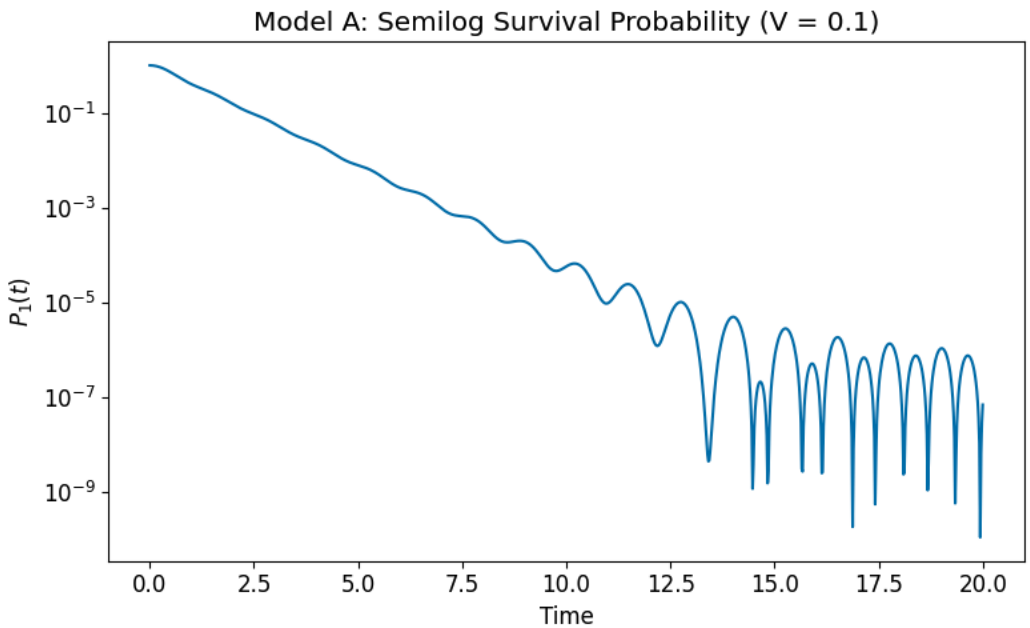

Figure 2. Semilogarithmic plot of the survival probability $P_1(t)$ for $V = 0.1$, showing approximate exponential decay at intermediate times.

The semilog plot (**Figure 2**) exhibits an approximately linear region at early to intermediate times, indicating that the decay follows an exponential form. At longer times, deviations from linearity and small oscillations appear due to finite-size effects associated with the discrete representation of the continuum.

4.2 Dependence of Relaxation on Coupling Strength

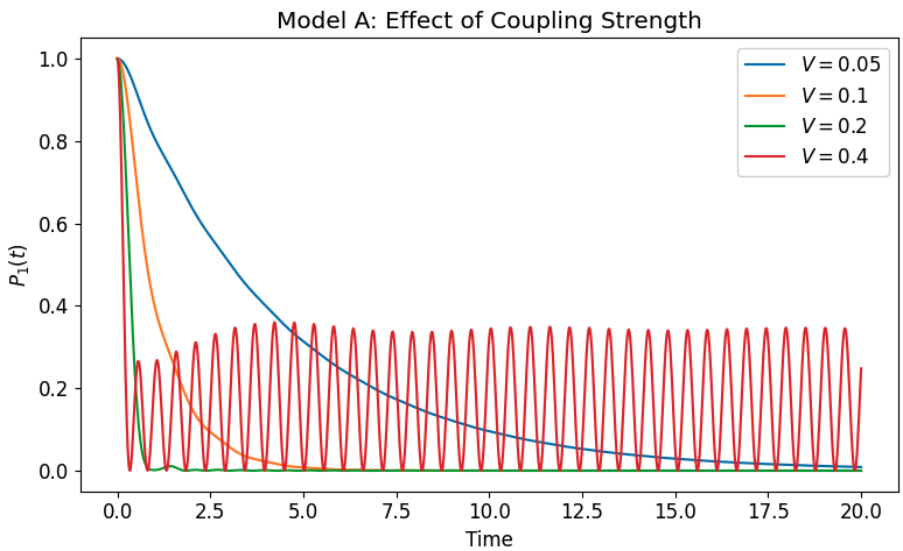

Figure 3. Survival probability $P_1(t)$ for different coupling strengths $V = 0.05, 0.1, 0.2, 0.4$, with $N = 150$ and $\Delta E = 10$.

Shown in **Figure 3** above, increasing the coupling strength V leads to a significantly faster decay of the survival probability. For weak coupling ($V = 0.05$), the decay is slow and smooth, while for moderate coupling ($V = 0.1$) the decay remains approximately exponential. At larger values ($V = 0.2$ and $V = 0.4$), the dynamics become strongly oscillatory, indicating a breakdown of the simple exponential relaxation regime.

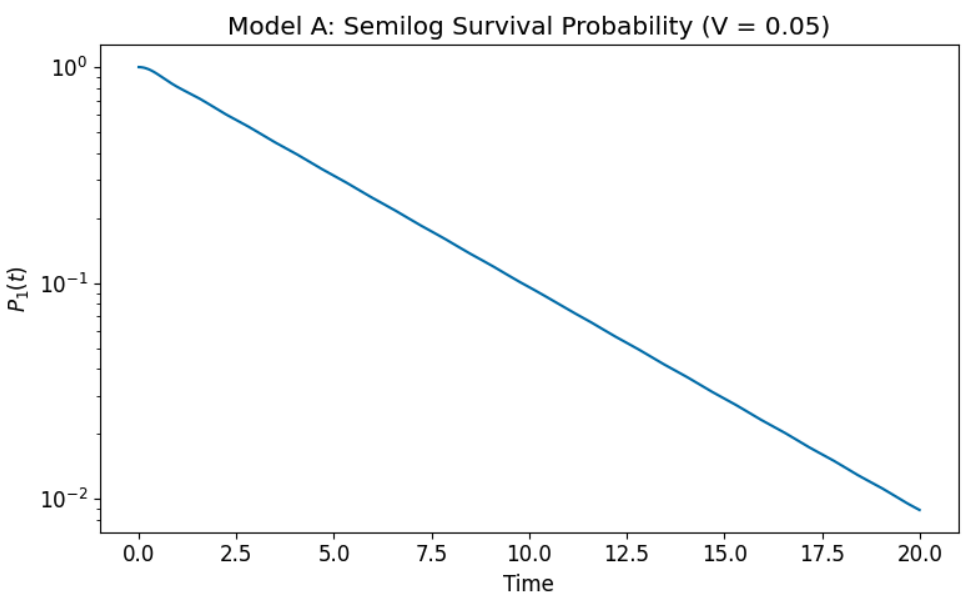

Figure 4. Semilogarithmic plot of $P_1(t)$ for $V = 0.05$, showing nearly perfect exponential decay

For $V = 0.05$, the semilog plot (**Figure 4**) is nearly perfectly linear over the entire time range, indicating ideal exponential decay. This behavior is characteristic of the weak-coupling regime, where the system effectively interacts with a dense continuum and relaxation follows Fermi's Golden Rule.

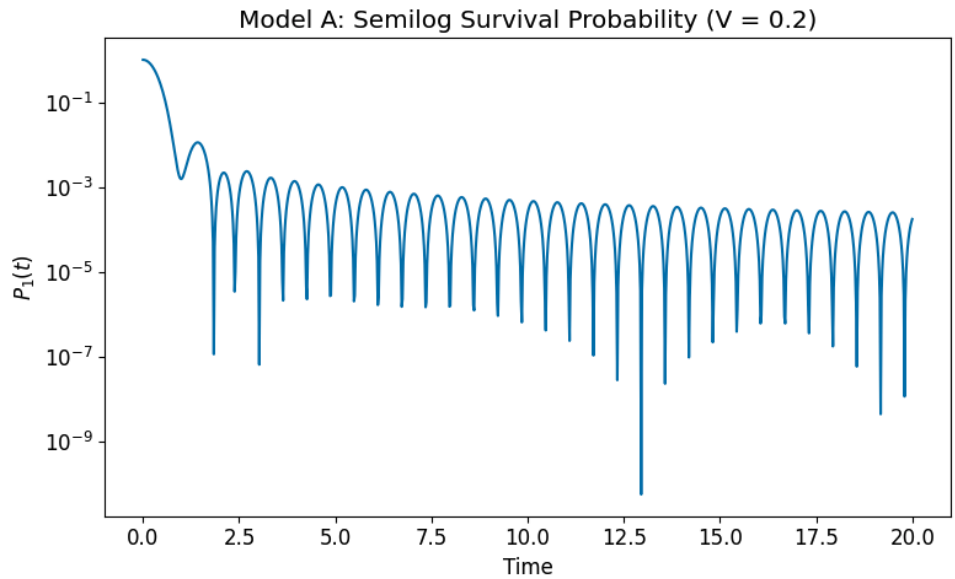

Figure 5. Semilogarithmic plot of $P_1(t)$ for $V = 0.2$, showing strong deviations from exponential decay

For large coupling strength ($V = 0.2$), the semilog plot (**Figure 5**) is highly non-linear and exhibits pronounced oscillations. This indicates that the system is no longer in the weak-coupling regime, and the dynamics are dominated by other interactions rather than irreversible relaxation.

4.3 Influence of Continuum Discretization

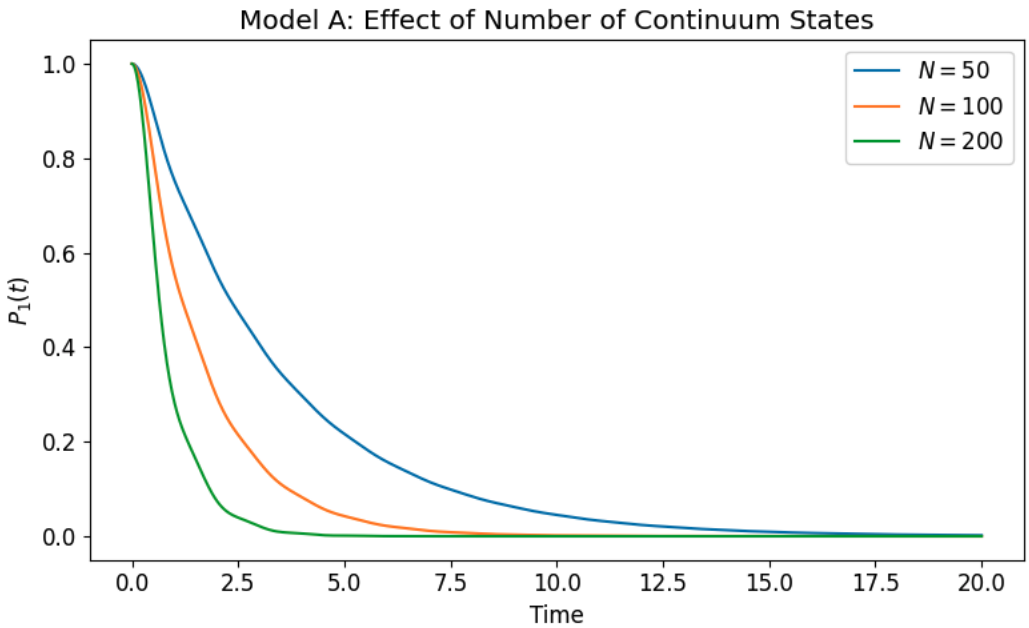

Figure 6. Survival probability $P_1(t)$ for different numbers of continuum states $N = 50, 100, 200$, with $\Delta E = 10$ and $V = 0.1$.

As shown in **Figure 6**, increasing the number of continuum states leads to a smoother and more rapid decay of the survival probability. For smaller N, the decay is slower. As N increases, the continuum becomes more densely populated, and the dynamics approaches those of a true continuum.

4.4 Dependence on Continuum Bandwidth

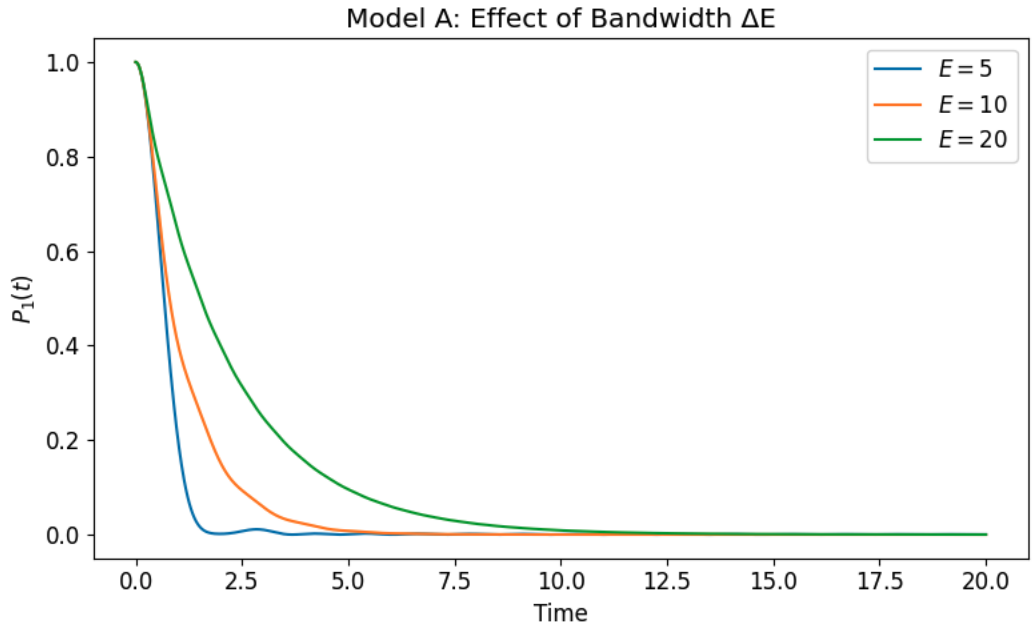

Figure 7. Survival probability $P_1(t)$ for different continuum bandwidths $\Delta E = 5, 10, 20$, with $N = 150$ and $V = 0.1$.

The bandwidth dependence must be interpreted together with the associated change in density of states. In the present calculations, N is held fixed while ΔE is varied, so decreasing ΔE reduces the level spacing and increases density of states $\rho = \frac{N}{\Delta E}$. As a result, smaller ΔE produces a faster decay of the survival probability $P_1(t)$, as shown in **Figure 7** above. However, the narrower continuum also introduces more visible structure in time dependence, indicating stronger finite-bandwidth effects. By contrast, larger ΔE gives a broader but less dense continuum, leading to slower and smoother decay.

4.5 Effect of Coupling Disorder

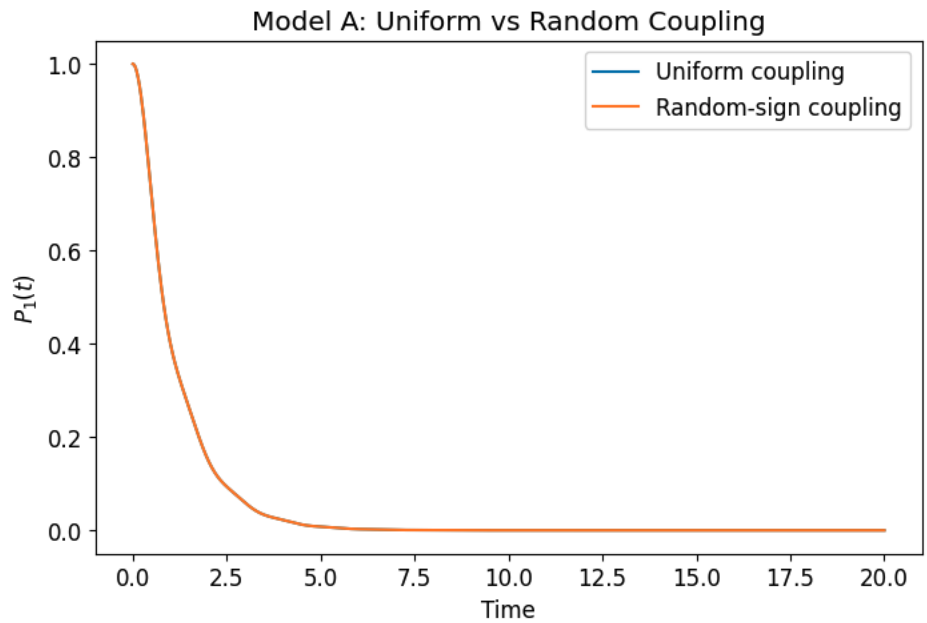

Figure 8. Comparison of survival probability $P_1(t)$ for uniform coupling $V_{1k} = V$ and random-sign coupling $V_{1k} = \pm V$, with identical $V = 0.1$

The survival probabilities for uniform and random-sign couplings in **Figure 8** are nearly identical when the magnitude of the coupling is fixed. This indicates that the relaxation dynamics depend primarily on the average coupling strength rather than the detailed sign structure of the coupling elements. The decay is therefore governed by statistical properties such as $\langle |V_{ik}|^2 \rangle$, rather than the specific configuration of coupling.

Overall, the results for Model (A) demonstrate that exponential relaxation emerges in an intermediate regime characterized by moderate coupling strength and a sufficiently dense continuum. The decay rate increases with coupling strength and with the density of continuum states, while deviations from exponential behavior arise when the coupling becomes too strong or when the continuum is insufficiently well approximated.

5. Results for Model (B)

5.1 Time Evolution of State Populations

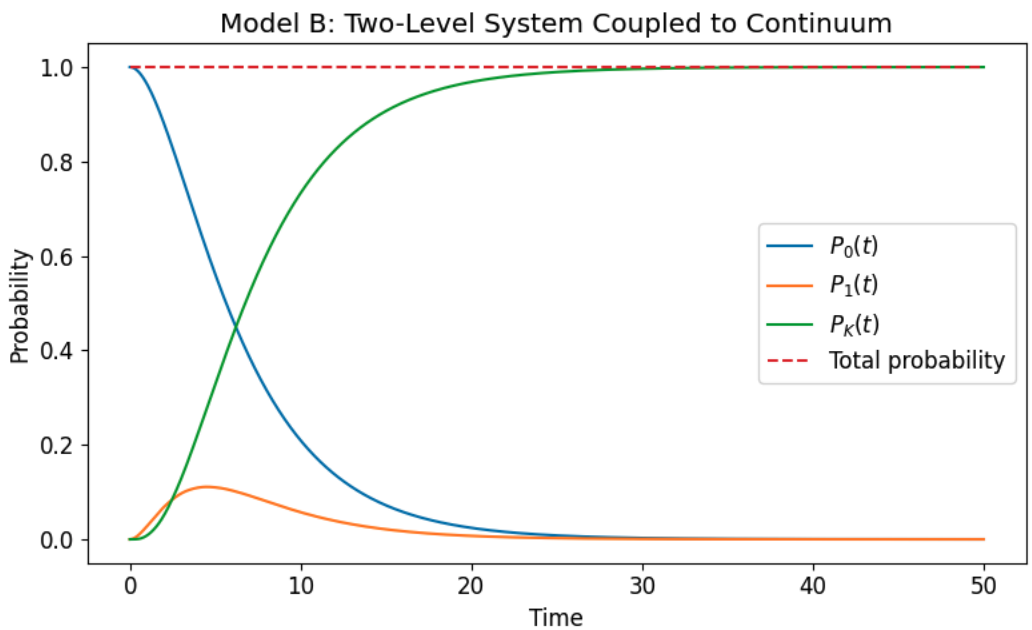

Figure 9. Time evolution of the probabilities $P_0(t)$, $P_1(t)$, and $P_k(t)$ for Model (B) with $N = 150$, $\Delta E = 10$, $V_{01} = 0.2$, and $V_{1k} = 0.1$. The total probability is also shown to verify normalization.

As demonstrated in **Figure 9**, the system initially occupies the state $|0\rangle$, and the probability $P_0(t)$ decreases over time as population is transferred to the intermediate state $|1\rangle$, and subsequently to the continuum. The population $P_1(t)$ exhibits a transient peak at early times, indicating temporary occupation of the intermediate before decaying. The continuum population $P_k(t)$ increases monotonically and eventually dominates, demonstrating sequential relaxation from $|0\rangle$ to $|1\rangle$ and then into the continuum. The total probability remains conserved throughout the evolution.

5.2 Dependence on Discrete State Coupling

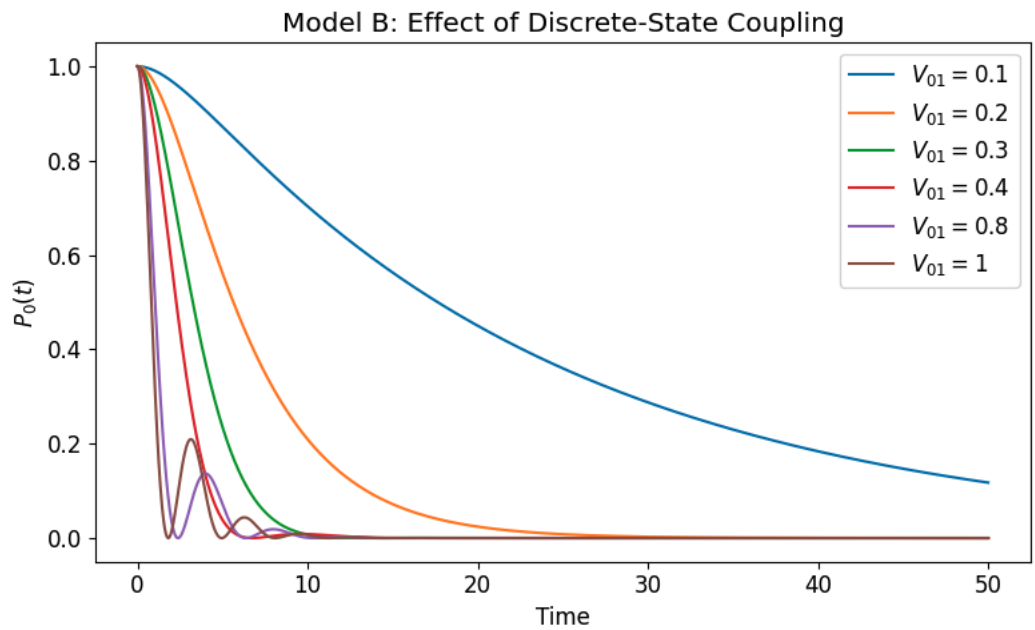

Figure 10. Survival probability $P_0(t)$ for different discrete state coupling strengths V_{01}, with $N = 150$, $\Delta E = 10$, and $V_{1k} = 0.1$.

Increasing the coupling V_{01} significantly accelerates the depletion of the initial state $|0\rangle$, as shown in **Figure 10** above. For smaller V_{01}, the decay is slow and smooth, indicating weak transfer to the intermediate state. As V_{01} increases, the decay becomes faster and more structured, with visible oscillations at larger values reflecting coherent population exchange between $|0\rangle$ and $|1\rangle$. These results show that V_{01} controls the rate and coherence of the initial transfer process.

5.3 Dependence on Continuum Coupling V_{1k}

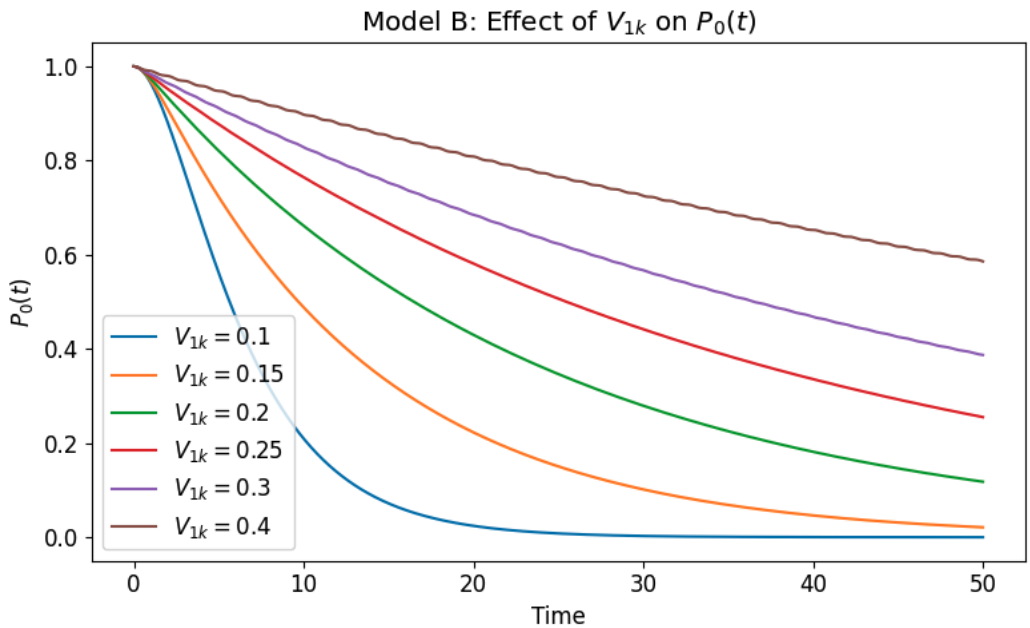

Figure 11. Time evolution of $P_0(t)$ for different continuum coupling strengths V_{1k}, with $N = 150$, $\Delta E = 10$, and $V_{10} = 0.2$

Shown in **Figure 11**, the dependence of $P_0(t)$ on V_{1k} is non-monotonic. For smaller V_{1k}, the decay of $P_0(t)$ is faster, while increasing V_{1k} leads to a slower and more gradual decay. This occurs because V_{1k} controls the lifetime of the intermediate state $|1\rangle$. When V_{1k} is small, population can accumulate in $|1\rangle$, facilitating efficient transfer from $|0\rangle$. In contrast, for large V_{1k}, the intermediate state rapidly loses population to the continuum, suppressing transfer from $|0\rangle$ and resulting in slower depletion.

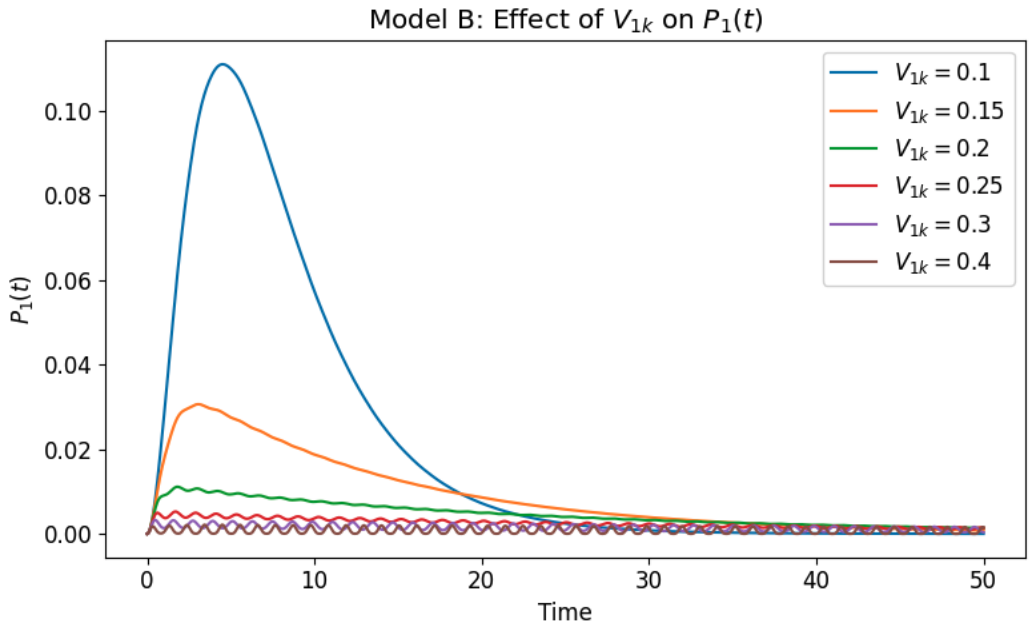

Figure 12. Time evolution of $P_1(t)$ for different continuum coupling strength V_{1k}

The above **Figure 12** show that the population of the intermediate state $P_1(t)$ decreases significantly as V_{1k} increases. For small V_{1k}, $P_1(t)$ reaches a relatively large peak, indicating substantial temporary population buildup. As V_{1k} increases, this peak becomes smaller and broader, reflecting the shorter lifetime of the intermediate state. At large V_{1k}, the population of $|1\rangle$ remains small at all times and oscillates because it rapidly decays into the continuum.

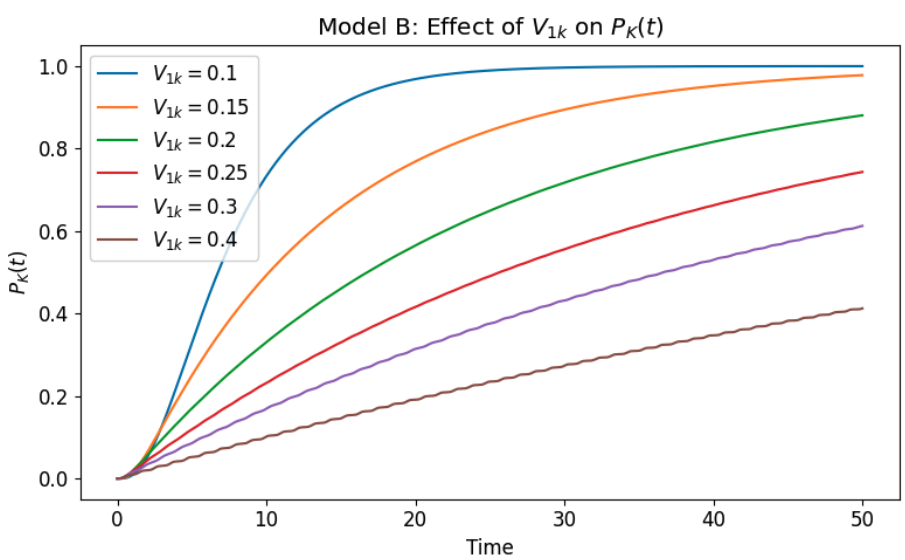

Figure 13. Time evolution of the continuum population $P_k(t)$ for different continuum coupling strengths V_{1k}

The dependence of $P_k(t)$ on V_{1k} is again non-monotonic (**Figure 13**). For smaller V_{1k}, the continuum population reaches 1.0 more rapidly, indicating efficient transfer through the intermediate state $|1\rangle$. In this regime, $|1\rangle$ has longer lifetime, allowing population to accumulate and be transferred effectively into the continuum. As V_{1k} increases, the growth of $P_k(t)$ becomes slower and more linear. This occurs because strong coupling to the continuum shortens the lifetime of $|1\rangle$, suppressing coherent population transfer from $|0\rangle$ and reducing the efficiency of the overall relaxation pathway.

These results demonstrate that the relaxation dynamics are governed by a competition between population transfer into the intermediate state and its decay into the continuum.

6. Conclusion

This project used numerical simulations to examine quantum relaxation in systems where discrete states are coupled to a continuum. Model (A) exhibits exponential decay in an intermediate regime, with deviations arising from finite-size effects and strong coupling. Model (B) reveals more complex dynamics due to the presence of an intermediate state, where relaxation depends on a balance between population transfer and decay. In particular, the dependence on continuum coupling is non-monotonic, highlighting the importance of intermediate-state lifetime. These results show that relaxation behavior is determined by both coupling strength and system structure.